

Hedge fund activism, R&D efficiency and firm value[☆]Justin Watson, Janice How^{*}, Peter Verhoeven

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ABSTRACT

We investigate the effectiveness of hedge fund activism in addressing inefficient R&D investment and creating long-term value at target firms. Using an endogenous growth model to estimate efficient R&D investment levels, we find target firms exhibit a tendency to overinvest in R&D. The likelihood of being targeted and activism announcement returns increase with R&D overinvestment. Despite observable cuts in R&D expenditure following activism, our analysis shows no significant improvement in R&D efficiency over the subsequent three years. Additional tests, controlling for the activist's objectives, reputation, and stock selection ability, reveal a negative impact of activism on value creation. Our study offers important insights into how hedge fund activism shapes corporate R&D strategies and highlights challenges in achieving sustainable innovation and value enhancement through activist interventions.

1. Introduction

The increasing prevalence of “investment-limiting” interventions by hedge fund activists has raised concerns about managerial ability to innovate efficiently.¹ The extant literature suggests that inherent market frictions often drive managers to engage in excessive investment (Bebchuk et al., 2015; Coffee and Palia, 2016), with hedge fund activism seen as a corrective measure guiding firms toward more efficient investment levels, thereby enhancing profitability and firm value (Brav et al., 2008; Greenwood and Schor, 2009; Bebchuk et al., 2015). Brav et al. (2018) document that although inputs to innovation, measured by R&D expenditure, decrease in absolute amounts during the five-year post-activism period, the quality and quantity of innovation outputs, such as patents, show an improvement. This positive outcome is attributed to the reallocation of innovative resources, redeployment of human capital, and enhancements in board-level expertise. However, despite these gains, target firms do not experience significant improvements and, in some cases, encounter no change or negative shifts in long-run stock performance following activism (Coffee and Palia, 2016; Cremers

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¹ Hedge fund activism has become synonymous with asset divestments (Klein and Zur, 2009; Chen and Feldman, 2018), board representation and CEO dismissals (Brav et al., 2008), significant cuts to investments and R&D expenditure (Bebchuk et al., 2015; Brav et al., 2018; DesJardine and Durand, 2020), and increases in leverage, payouts and profitability (Brav et al., 2008; Klein and Zur, 2009; Bebchuk et al., 2015).

et al., 2020; DesJardine and Durand, 2020).²

This paper re-examines whether activists strategically target and optimize firms characterized by overinvestment in R&D by estimating the “efficient” level of R&D investment, representing the point where the firm maximizes its innovative capacity given available resources. We focus on R&D expenditure as it is widely recognized as the firm’s primary means of corporate rejuvenation, growth, development, and long-term survival (Covin and Miles, 1999; Sørensen and Stuart, 2000).

Our approach is two-fold. First, we establish a theoretical framework linking efficient R&D investment, R&D productivity, and firm value. Consistent with endogenous growth theory (Romer, 1990), the firm’s production is a non-linear function of technology, which increases in R&D expenditure, and a generic input, which is designed to capture other relevant production factors. This framework allows us to capture growth and efficiency channels to profit and value creation, both of which are central to the polarizing hedge fund activism literature.³ We evaluate this relation in a standard neo-classical framework and employ a random-coefficients regression to estimate firm-level R&D productivity (Knott, 2008). Second, based on this framework, we derive firm-level estimates of R&D efficiency and measure R&D inefficiency as deviations from this level. Our model relies primarily on a firm’s R&D productivity, incorporating key elements such as existing technology and its depreciation rate, the level of capital, labour and advertising, and productivity of these inputs, in line with prior endogenous growth constructs (Thompson, 1996; Knott, 2008).

Based on a sample of 636 activism events at target firms that disclosed R&D expenditure at least once⁴ during the period 1998–2011, we present the following key results. First, we find a substantial 73 percent of target firms exhibit R&D overinvestment, averaging \$12.4 million or 6.3 percent of market capitalization. This finding supports the prevailing view that market frictions bias managers towards excessive investment (Bebchuk et al., 2015; Coffee and Palia, 2016). Second, our instrumental variable analysis shows the likelihood of becoming a target for activists increases with R&D overinvestment, suggesting that activists may well have intended to address managerial inefficiencies (Brav et al., 2008; Greenwood and Schor, 2009; Bebchuk et al., 2015). Third, activism announcement returns are positively related to R&D overinvestment and negatively related to R&D underinvestment. Given activists’ tendency to reduce R&D expenditure in target firms (Brav et al., 2018), this finding suggests that the market perceives activism as value-enhancing in overinvesting firms, but value-destroying in underinvesting firms.

Our difference-in-difference (DiD) analysis shows that, despite observed reductions in R&D expenditure, target firms experience no significant improvements in the efficiency of R&D investment in the three years following activism. Analysis of factors of production reveals that the cut in R&D expenditure is offset by reductions in the scale of the firm, resulting in minor changes in inefficient R&D investment, long-term shareholder value, and operational performance. Our results on long-term activism outcomes remain robust when we consider activists’ stated objectives, reputation, and ownership stake in our analysis, and when we assess the treatment effects of switches in an activist’s stance from passive to active.

In summary, our findings suggest that activists make firms leaner in a way that neither addresses inefficient R&D investment nor enhances long-term value, thus adding to the puzzle why activist investors target firms that do not inherently offer long-term value (Coffee and Palia, 2016; Cremers et al., 2020; DesJardine and Durand, 2020). Moon (2024) shows that financial conditions influence R&D choices, with financially strained firms opting for high-risk, high-reward projects, while better-funded firms choose safer, modest-return projects. This pattern suggests that hedge fund activism might drive firms towards riskier R&D for short-term gains, potentially compromising long-term innovation and growth. Alternatively, our results might stem from activists’ inability to identify R&D investment inefficiencies at target firms, given the high level of information asymmetry surrounding R&D expenditures and innovation-related activities.

Our paper contributes to the ongoing discourse on the value implications of hedge fund activism, particularly in the context of R&D efficiency and long-term value creation. First, we introduce and apply a novel measure of efficient R&D investment at target firms based on managerial capabilities. Unlike existing literature that largely infers R&D efficiency through targeting behavior and activism outcomes (Brav et al., 2010, 2018), our approach provides a direct measure. This contribution is crucial given the polarizing debates surrounding the concept of “efficient investment” in the hedge fund activism literature (Coffee and Palia, 2016; Gantchev et al., 2020).

Our study sets itself apart from Brav et al. (2018) by shifting the focus from the general influence of hedge fund activism on corporate innovation to a more specific inquiry into whether such activism corrects inefficient R&D investments and contributes to long-term value creation in target firms. Despite observed reductions in R&D expenditure post-activism, we find no significant improvements in R&D efficiency, challenging the optimistic portrayal of activists as efficiency-enhancing agents (Brav et al., 2008, 2018; Greenwood and Schor, 2009; Bebchuk et al., 2015). Our findings thus contribute a nuanced understanding to the broader discourse on the governance role of activists in shaping corporate innovation strategies.

Our contribution also lies in our empirical approach, which explicitly controls for the nature of missing accounting data (Koh and Reeb, 2015) and the complex interrelationships between firm inputs in the innovation process (Crossan and Apaydin, 2010); the latter forms part of a growing literature that utilizes random-coefficients models in production function estimation (Knott, 2008; Doraszelski

² Firms that receive a takeover offer following an activist intervention are an exception. For these firms, the long-run stock price performance improves (Greenwood and Schor, 2009; Boyson and Pichler, 2019).

³ While one strand of the literature argues that activist interventions add value by promoting more efficient investment practices (Bebchuk et al., 2015; Brav et al., 2008, 2018), the other strand suggests that hedge fund activism constitutes asset-stripping and hinders a target’s ability to grow and create value in the long-term (Coffee and Palia, 2016; Cremers et al., 2020; DesJardine and Durand, 2020).

⁴ Brav et al. (2018) impose a more stringent criterion, requiring firms to report R&D at least once in the previous five years in addition to having lodged a patent any time prior to activism. We do not consider patenting activity because our focus is on the firm’s factors of production and its ability to convert these into revenue (i.e., productivity).

and Jaumandreu, 2013; 2018). With these methodological approaches, we provide new insights into how activists select their targets, how the market initially perceives the success of the activist's campaign, and the somewhat muted influence of hedge fund activism on long-run value creation in target firms.

Our research offers practical insights into the links between activism, R&D efficiency, and firm value. Investors gain crucial insights, discerning varied outcomes based on target firms' initial R&D investment status. For corporate decision-makers, especially those vulnerable to activist actions, grasping this relationship is vital. Despite potential short-term gains, activist interventions often fail to address underlying R&D investment inefficiencies, raising concerns about sustained innovation and value creation. Policy-makers can leverage our insights to craft policies that promote innovation-driven economic growth amidst increased regulatory scrutiny of hedge fund activism.

Our findings also highlight opportunities for corporate governance reforms aimed at addressing R&D investment inefficiencies and fostering transparency and accountability within firms. Evidence from Chinese firms shows that over-investment in R&D leads to lower bank loan financing sensitivity and higher trade credit financing sensitivity due to managerial expropriation of resources (Chiu et al., 2024). This phenomenon underscores the broader implications of inefficient R&D investment, emphasizing the need for policies that balance short-term gains with long-term innovation objectives. Understanding the complex relationship between activism and innovation efficiency is crucial for fostering an environment conducive to sustainable economic development and innovation.

2. R&D efficiency

2.1. Theoretical framework

Drawing on Schumpeter's (1934) idea that innovation is the setting up of a new production function, we examine the firm in a standard neo-classical economic framework by modelling its production function and ability to create output through the utilization of inputs. We recognize innovation as a multi-faceted problem with complex interrelations between various factors in the firm's innovation process. To model the dynamics of the firm's investment strategy, we consider a multi-product firm whose products are assumed to be imperfect substitutes for other products in the market. In this sense, the market is not perfectly competitive, with the firm being able to price its products above marginal cost to generate profit. Without loss of generality, the firm has an existing base of inputs that it can utilise to generate output. The firm derives a portion of its output (and therefore profitability) from its technology stock, which reflects the firm's R&D expenditure over time.

The firm's managers are assumed to be risk-neutral and rational agents who seek to maximize firm value through their choice of R&D expenditure, $R_{i,t}$, in the production of the firm's output.⁵ Assuming Cobb-Douglas technology, the firm's production function of output, Y , represents a combination of two inputs: technology (T) and a generic input (X) designed to capture other factors relevant to the production process:

$$Y_{i,t} = AT_{i,t}^{\gamma_i} X_{i,t}^{\theta_i} \quad (1)$$

where $0 < \gamma_i < 1$, $0 < \theta_i < 1$. Here, A represents total factor productivity (or the portion of output not explained by the model's inputs); γ represents the output elasticity of technology; and θ represents the output elasticity of the firm's generic inputs. Both γ and θ are a function of the firm's capacity to convert its investments into workable inputs (or production technologies) and to utilize its production technologies to generate output. The firm's technology stock accumulates over time from investment in R&D, R :

$$T_{i,t} = (1 - \delta_{i,t})T_{i,t-1} + R_{i,t} \quad (2)$$

where $\delta_{i,t}$ represents the depreciation rate on previous R&D expenditure such that the technology stock at any given point in time is the sum of all incremental R&D expenditure depreciated up until time t : $T_{i,t} = \sum_{T=1}^t \delta_i^T R_{i,T}$. Thus, the firm continues to generate output from its previous investment in R&D, although the incremental contribution of technology to the firm's output stream depreciates over time. Since the firm's technology stock depends solely on R&D expenditure, γ_i represents the firm's R&D productivity, i.e., how successful the firm is at converting R&D expenditure into workable output.

Profitability is simply the firm's output less the cost of inputs. The cost per unit of X is w and the cost of incremental R&D expenditure is 1, so R captures actual R&D expenditure.⁶ Therefore, firm profit ($\pi_{i,t}$) is:

$$\pi_{i,t} = AT_{i,t}^{\gamma_i} X_{i,t}^{\theta_i} - R_{i,t} - wX_{i,t} \quad (3)$$

$$\pi_{i,t} = A(T_{i,t-1} + R_{i,t})^{\gamma_i} X_{i,t}^{\theta_i} - R_{i,t} - wX_{i,t}$$

⁵ The risk preferences and rationality of managers are more relevant to the question of *why* managers may be investing inefficiently in R&D, rather than how much managers *should* be investing in R&D. In any case, if managers are investing inefficiently, it provides motivation for hedge fund activists to target the firm.

⁶ The price per unit of X , w , is intended to be generic. For some inputs such as R&D and advertising, the price per unit is 1 since they are denominated in dollars and are independent of the firm's existing technological or reputational capital. For inputs such as capital and labour, the cost is proportional to existing capital, thus the price typically reflects the cost of capital or, in the case of labour, mean wage.

Consistent with Thompson (1996), we allow the independent values to vary over time. However, we assume steady-state expectations such that there exists, across all market participants, a homogenous expectation at time t that the values remain constant in perpetuity. This assumption maintains a workable relation between R&D expenditure and market value, which is not overcomplicated by continuous time mathematics and dynamic modelling. In addition, the discrete-time model matches the nature of accounting data and the flow and incorporation of relevant information into market prices.

Assuming steady state expectations, the market value of the firm ($V_{i,t}$) is the present value of future profits:

$$V_{i,t} = \sum_{T=t}^{\infty} \varphi^{T-t} \left[A_i \left(T_{i,t} + \sum_{T=t}^{\infty} \delta_{i,t} R_{i,t} \right)^{\gamma_i} X_{i,t}^{\theta_i} - R_{i,t} - wX_{i,t} \right] \quad (4)$$

with discount factor $\varphi = \frac{1}{(1+\rho)}$, where ρ is the risk-free rate. Efficient levels of R and X are denoted by R^* and X^* , respectively. Since we assume the expected values of R^* and X^* are time invariant, $R_{i,t}^* = R_i^*$ and $X_{i,t}^* = X_i^*$. Substituting into Eq. (4) and summing the geometric series results in:

$$V_{i,t} = \underbrace{\frac{1}{\rho} \left[A_i \left(T_{i,t} + \frac{R_i^*}{\delta} \right)^{\gamma_i} X_i^{\theta_i} \right]}_{\text{Present value of revenues}} - \underbrace{\frac{1}{\rho} [R_i^* + wX_i^*]}_{\text{Present value of costs}} \quad (5)$$

Eq. (5) posits that firm value decreases by the present value of expected future R&D expenditure and increases by the present value of R&D's expected incremental contribution to the firm's technology stock, conditional on the firm's R&D productivity, γ_i . Since the firm's objective is to maximize $V_{i,t}$ by choice of inputs, the efficient level of R_i^* can be found by taking the partial derivative of $V_{i,t}$ with respect to R and setting it to zero. Hence, we make the following proposition:

Proposition. *There exists a level of efficient R&D expenditure (R_i^*), which is dependent on the firm's other production inputs ($X_i^{\theta_i}$), existing technology stock (T_i), R&D productivity (γ_i), and the depreciation (discount) rate on technology (δ):*

$$R_i^* = \left[\left(\frac{\delta}{A_i \gamma_i X_i^{\theta_i}} \right)^{\frac{1}{\gamma_i-1}} - T_{i,t} \right] \times \delta \quad (6)$$

$$X_i^* = \left(\frac{w}{A_i \theta_i T_i^{\gamma_i}} \right)^{\frac{1}{\theta_i-1}} \quad (7)$$

The incremental contribution of R&D expenditure to firm value exhibits decreasing returns to scale (as long as $0 < \gamma < 1$), while R&D expenditure itself decreases firm value in a linear fashion (since the price of incremental R&D expenditure is 1).

2.2. Empirical estimation

To identify the efficient level of R&D investment, we would first require an estimation of the firm's production function to obtain firm-specific estimates of productivity. We estimate the firm's production function using the merged Centre for Research on Security Prices (CRSP)-Compustat database for the period 1985–2015. Consistent with standard practice (Montresor and Vezzani, 2015), we augment Eq. (1) to include the firm's technology, capital, labour, and advertising:

$$Y = AK_{it}^{\phi} L_{it}^{\beta} T_{it}^{\gamma} D_{it}^{\psi} e^{\epsilon_{it}} \quad (8)$$

where Y is the firm's output; K is capital; L is labour; T is technology; D is advertising; and ϵ is the error term. In Eq. (8), Y is empirically measured by the firm's revenue; K by net property, plant and equipment (PPE); L by the number of full-time employees; and D by advertising expenditure. The firm's technology stock (T) is computed as the sum of all R&D expenditures depreciated over time: $T_{i,t} = \sum_{T=1}^t \delta_{i,t}^{T-t} R_{i,T}$. Following prior studies (Montresor and Vezzani, 2015), we use 15 percent as the rate of depreciation of technology. To identify the level of efficient R&D investment, we require the output elasticities of all the firm's inputs: ϕ , β , γ , and ψ .

Taking the logarithm of Eq. (8), we obtain:

$$y_{it} = (\alpha + \alpha_i) + (\phi + \phi_i)k_{it} + (\beta + \beta_i)l_{it} + (\gamma + \gamma_i)t_{it-1} + (\psi + \psi_i)d_{it} + \epsilon_{i,t} \quad (9)$$

where small letters denote the variables defined as above but in their logarithmic form. Eq. (9) is estimated using a random-coefficients regression (RCR) model and maximum likelihood (ML), modelling heterogeneity at both the industry and firm levels using Fama-French 12-industry classifications.⁷ Consistent with Knott (2008), we estimate Eq. (9) for each firm-year with 10-year rolling windows from 1985 to 2015. This method involves using a lag of the firm's technology stock to mitigate endogeneity concerns arising from delays between a firm's R&D expenditure and the subsequent payoffs. Thus, the first year of usable data is 1995, three years before the first activism event.

A concern in the estimation of the production function is potential correlations between inputs and unobserved firm-specific shocks in productivity. Since R&D expenditure is unlikely to generate outputs immediately, R&D expenditure should only respond to stable firm-specific shocks captured by the firm-specific error term (Knott, 2008).

Our final step is the estimation of the efficient level of R&D investment for each firm. We expand Eq. (6), which expresses the general functional form of the firm's level of efficient R&D investment, to include the firm's observable inputs as follows:

$$R_{it}^* = \left[\left(\frac{\delta}{A_{it}\gamma_{it}K_{it}^{\phi_{it}}L_{it}^{\beta_{it}}D_{it}^{\psi_{it}}} \right)^{\frac{1}{\gamma_{it}-1}} - T_{it} \right] \times \delta \quad (10)$$

where the variables are as defined earlier, and the coefficients are those estimated using the RCR methodology. The inefficient level of R&D investment for firm i at time t is computed as the absolute difference between the actual R&D expenditure and the estimated efficient level of R&D investment, i.e., $|(R_{it} - R_{it}^*)|$.

3. Data and research method

3.1. Data

Our initial dataset, which consists of 2684 hedge fund activist interventions from 1998 through 2011, was obtained from Alon Brav.⁸ Since our focus is on R&D firms, we remove 2048 interventions that do not belong to the research-active sectors of manufacturing, technology, and healthcare and that do not report R&D activity. This selection procedure results in a final sample of 636 target firms. Data pertaining to company fundamentals are sourced from Compustat and stock price data are obtained from the CRSP database.

3.2. Research method

3.2.1. The likelihood of hedge fund activism and R&D inefficiency

If activist hedge funds are motivated to discipline managerial inefficiencies, we expect to find a positive relation between a firm's level of inefficient R&D investment (measured in absolute terms) and the likelihood of the firm being targeted. The greater incentive to target firms with higher levels of inefficient R&D investment is also driven by hedge funds' desire to offset the cost of activism.⁹ To test this prediction, we use a two-stage least squares (2SLS) fixed effect regression model to account for potential endogeneity and unobserved heterogeneity in our estimates of inefficient R&D investment.

In the first-stage regression, we instrument *R&D Inefficiency* with the natural logarithm of *Revenue* and *Assets*:

$$R\&DInefficiency_{it} = \beta_0 + \beta_1 Revenue_{it} + \beta_2 Assets_{it} + \gamma_i x'_{it} + \varepsilon_{it} \quad (11)$$

By design, *R&D Inefficiency* is related to firm output and input, which are correlated to revenue and assets. Both instruments meet

⁷ The RCR model offers several advantages relative to other approaches to production function estimation. First, it treats coefficients as non-fixed and potentially correlated with the error term (Knott, 2008). Traditional approaches such as ordinary least squares (OLS) and generalised methods of moments (GMM) typically assume fixed coefficients, implying a stable production process across firms and that firms use the same technology (Montesoro and Vezzani, 2015). Second, the RCR model generates best linear unbiased predictors of firm-specific productivity (Swamy and Tavlak, 1995; Knott, 2008), allowing us to directly observe the heterogeneity exhibited by firms in the utilisation of inputs. Finally, the RCR model explicitly deals with simultaneity in the estimation of production functions.

⁸ As outlined by Brav et al. (2008), the list of activism events (Schedule 13D filings) and amendments (Schedule 13D/A filings) was extracted from the SEC's Electronic Data Gathering, Analysis and Retrieval (EDGAR) database. To mitigate the concern that Schedule 13D filings are biased towards small-cap targets (easier to meet the five percent threshold), media outlets and Schedule 13 F filings were searched for evidence of activism in large-cap firms. Announcements where the primary purpose of the intervention is risk arbitrage or distress financing, and filings from closed-end funds were excluded.

⁹ Cost of activism include the hiring of professional advisors, the development of reports, the cost of shareholder engagement and coordinating and unifying a dispersed ownership base, and other intangible costs such as loss of reputation should the campaign fail (Brav et al., 2008; Gantchev, 2013; Johnson and Swem, 2021).

the exclusion restriction as they are unlikely to be associated with the targeting decision for at least two reasons. First, performance incentives would require activists to pursue channels that generate positive abnormal returns for their investors (Fung et al., 2008). Thus, when deciding on a target firm, rather than focusing on the level of assets and revenue, activists are more inclined to focus on whether the firm's resources are being utilized efficiently, whether assets and growth opportunities are undervalued, and the means through which value can be extracted (Brav et al., 2008, 2010; Klein and Zur, 2009; Coffee and Palia, 2016). Second, activists would consider the practicality of acquiring an influential ownership position and whether they can push their agenda in their targeting decision (Brav et al., 2008; Coffee and Palia, 2016). This consideration substantiates why the firm's market capitalization and position in the industry drive the targeting decision rather than revenue and assets.

In the second stage, we run a linear probability model of the following general form:

$$Target_{i,t} = \beta_0 + \beta_1 R\&DInefficiency_{i,t} + \gamma_i x'_{i,t} + \varepsilon_{i,t} \quad (12)$$

where *Target* is a dummy that equals one if the firm is targeted by activists, and zero otherwise. Our variable of interest is the instrumented *R&D Inefficiency*. In the above equation, $x'_{i,t}$ represents a vector of firm characteristics that are likely related to the targeting decision. Evidence shows activists target profitable, small-cap firms with poor stock performance (Brav et al., 2008; Klein and Zur, 2009; Cremers et al., 2020). Thus, we control for the rate of return on assets (*ROA*), market capitalization (*Market Cap*), *Tobin's Q*, and prior stock performance (*Return*). Activists are also more likely to target firms that encounter agency costs associated with free cash flow, as implied by high cash levels (*Cash*) and low sales growth (*Growth*), leverage (*Leverage*), and payouts (*Payout*) (Brav et al., 2010). We include the cost of equity (*CAPM*), book-to-market ratio (*B/M*),¹⁰ and market share (*Market Share*) to control for systematic risk, (mis)pricing, and industry saturation, respectively. Industry, year, and industry-year fixed effects are included to account for unobserved industry and macroeconomic effects. To reduce the adverse impact of outliers on the regression results, continuous variables are winsorized at the 1st and 99th percentiles.

3.2.2. Announcement returns around activism

To test the extent to which potential remediation of R&D inefficiency is reflected in stock returns around activism announcements, we run the following 2SLS fixed effect regression:

$$CAR_{i,t} = \beta_0 + \beta_1 R\&DInefficiency_{i,t} + \gamma_i x'_{i,t} + \varepsilon_{i,t} \quad (13)$$

where $CAR_{i,t}$ is the cumulative market-adjusted return computed over the 3-day (-1,+1) and 21-day (-10,+10) windows surrounding the activism announcement. The 3-day window coincides with the filing of Schedule 13D, which indicates the activist's intentions. We examine the 21-day window since activists are required to report their position within 10 days of crossing the five percent beneficial ownership threshold. Prior studies observe positive movements in stock price as activists acquire their positions during this longer window (Brav et al., 2008; Bebchuk et al., 2015).

We control for transaction type and hedge fund fixed effects because the market is likely to react to the activist's agenda and reputation for creating value than to firm-specific characteristics (Greenwood and Schor, 2009; Krishnan et al., 2016), which we also control for. Industry and year fixed effects and dummies for each of the general objective categories¹¹ are included. Since activism is costly, the decision to target a firm is likely to signal the activist's strategic evaluation that the benefits of mitigating R&D inefficiency exceed the costs of doing so. Further, traditional institutional investors, who prefer a focus on core products and services, are largely hampered by regulatory and structural barriers from becoming active to protect their interests (Brav et al., 2008). Our reasoning is consistent with the view that hedge funds are the natural champions of dispersed and diversified shareholders who are less capable of collective action in their own interest (Coffee and Palia, 2016). We expect a positive relation between inefficient R&D investment and announcement returns. As before, we instrument *R&D Inefficiency* using the natural logarithm of *Revenue* and *Assets*.

3.2.3. Hedge fund activism and changes in R&D inefficiency

To examine whether hedge fund activists facilitate more efficient R&D practices, we adopt the following difference-in-difference (DiD) regression framework:

$$y_{i,t} = \alpha_i + \beta_1 \bullet HFA_i \times Post_{i,t} + \beta_2 Post_{i,t} + \gamma_i x'_{i,t} + \varepsilon_{i,t} \quad (14)$$

We include observations for both target firms and matched control firms from three years prior to three years after the intervention. Target firms are propensity score-matched with non-targeted R&D firms on industry, activism year, market capitalization, book-to-market ratio, return on assets, market share, and *Tobin's Q* as well as the change in these variables from $t-3$ to $t-1$. Except for *PPE*, target and matched firms are indistinguishable in firm characteristics in the year prior to activism (untabulated), providing confidence about the quality of our matching process.

¹⁰ The correlation between *Tobin's Q* and *B/M* is -0.665 , and the variance inflation factor (VIF) is low (1.35), indicating that multicollinearity does not pose a significant issue in our analysis.

¹¹ The categories are general investment, capital structure, business strategy, sale, and governance (see Brav et al. 2008, Klein and Zur 2009, and Greenwood and Schor 2009).

3.2.4. Dealing with missing R&D values

To avoid biases resulting from conventional treatments of missing R&D data, which are replaced with either a zero or the industry average,¹² we impute the missing values using multiple imputation with chained predictive mean matching (MICE-PMM).¹³ Following standard practice, we include all test and auxiliary variables that are potentially correlated with the missing data in the imputation model (White et al., 2011). We estimate a probit regression for all firms with non-missing R&D (advertising). The dependent variable is one if the reported R&D (advertising) is “0”, and zero if it is non-zero.¹⁴ Using the estimated coefficients, predicted probabilities are generated for all firms (including missing R&D firms) to see whether firms with missing R&D are more likely to have zero R&D or non-zero R&D but not report it. The predicted probability is then included in the imputation method to help generate the missing R&D value. Following Morris et al., (2014), we generate five datasets ($M=5$), set the pool of nearest neighbours to 10, and use 20 iterations to achieve stable imputations. We also impute data at the industry level to maintain data structure across industries.

4. Summary statistics and empirical results

4.1. Summary statistics

Table 1 provides the frequency distribution of activism events by year and industry. Just over half (335 or 52.67 percent) of target firms belong to the technology sector. Healthcare and manufacturing firms make up 28.93 and 18.39 percent of the sample, respectively. We observe pro-cyclical behaviour in activism activities, consistent with Brav et al. (2008), with a higher frequency of activist interventions during economic booms (2005–2008) than in recessionary periods (pre-2005 and post-global financial crisis (GFC) periods).

Summary statistics in Table 2 show the average (median) dollar value of the signed *R&D Inefficiency* at target firms is \$12.3 million (\$5.38 million), representing 6.3 (4.6) percent of the firm's market capitalization. It has a positive sign, implying overinvestment in R&D. The proportion of target firms that overinvest in R&D is 73.1 percent (Panel A), with the highest proportion (85.9 percent) recorded for healthcare targets (Panel B). The proportion of target firms overinvesting in R&D has declined considerably over the sample period, from 91 percent in 1998–49 percent in 2009 (Panel C). The slight increase in the proportion of firms that overinvest in R&D in the two post-GFC years coincides with the “second wave” of hedge fund activism, which saw a shift from pump-and-dump activist strategies to more strategic and hands-on interventions (Krishnan et al., 2016).

Panel D shows target firms tend to be small, with an average (median) market capitalization of \$733 million (\$157.5 million) and an average workforce of 2238 employees. They also tend to have low leverage (15.8 percent), high growth (21.2 percent), a negative return on assets (-0.03 percent), and pay out on average just 2.2 percent of their earnings to their shareholders.

4.2. Empirical results

4.2.1. Do activists target firms that invest inefficiently in R&D?

Table 3 presents results from the IV-2SLS linear probability model, with standard errors clustered at the firm level. First-stage regression results for the full sample show *R&D Inefficiency*, measured in absolute terms (column (1)), is strongly related to its predictors, providing some validation of our theoretical framework.¹⁵ The coefficients on *Leverage*, *Payout*, *Market Cap*, *Return*, *Tobin's Q*, *Growth*, and *CAPM* are significant, suggesting that R&D inefficiency is associated with debt-overhang (Myers, 1977), empire-building (Jensen, 1986), industry dynamics (Fama, 1980), and shareholder risk preferences (Aghion et al., 2013). The *F*-statistics exceed 10 (Staiger and Stock, 1997) and Sargan's (1958) *J*-statistics are insignificant in all regressions; thus, the instruments meet both relevance and exogeneity requirements.

The second-stage results in Panel A show that for the full sample, both measures of *R&D Inefficiency* are positively related to the likelihood of being targeted (columns (2) and (5)). Therefore, activists are more likely to target firms with inefficient R&D investment, driven by a motivation to address managerial inefficiencies (Brav et al., 2008; Greenwood and Schor, 2009; Bebchuk et al., 2015) and offset activism cost. Overall, firms with higher profitability (ROA) and cash levels (*Cash*), and better prior stock performance (*Return*) are more likely to be targeted. Larger firms (*Market Cap*) and firms with higher payouts, asset turnover, market share, and systematic risk (*CAPM*) are less likely to be targeted, consistent with prior literature (Brav et al., 2008; Klein and Zur, 2009).

We explore a potential channel through which R&D inefficiency may influence activists' targeting behaviour, based on whether targets are overinvesting (columns (3) and (6)) or underinvesting (columns (4) and (7)) in R&D. Our results show a positive and

¹² Koh and Reeb (2015) note many firms with missing R&D data engage in innovation activities.

¹³ The MICE-PM method has the benefit of incorporating uncertainty about the true value of a variable and associations between variables in both the imputation procedure and empirical analysis. It also has the effect of making significant results harder to find, yet free from imputation bias. The procedure involves making linear predictions of the missing values with an imputation model, randomly drawing a value from a pool of observed values with the closest predicted value. Multiple datasets are generated and empirical analysis is performed on each dataset with coefficient estimates, standard errors, and test statistics pooled in accordance with Rubin's (1987) rules.

¹⁴ Since the rates of missingness in other accounting variables (such PPE and number of employees) are close to zero percent, we do not estimate probit regressions for those variables.

¹⁵ To save space, we report the first-stage regression results only for the full sample. Qualitatively similar first-stage regression results (untabulated) are found for *R&D Inefficiency* measured as a percentage of firm size.

Table 1
Frequency distribution of targeted firms by industry and year.

Year	Industry Sector			Total
	Manufacturing	Technology	Health	
1998	8	11	17	36
1999	7	18	2	27
2000	10	12	3	25
2001	7	13	3	23
2002	5	24	6	35
2003	7	19	9	35
2004	5	14	13	32
2005	17	28	13	58
2006	14	36	22	72
2007	9	47	21	77
2008	12	34	33	79
2009	1	17	15	33
2010	7	32	15	54
2011	8	30	12	50
Total	117	335	184	636

This table reports the frequency distribution of R&D firms targeted by hedge fund activists during the period 1998–2011 by year and industry (using the Fama & French 12-industry classification). An R&D firm is one that has reported R&D expenditure at least once during the sample period.

significant coefficient on *R&D Inefficiency* only for overinvesting targets (columns (3) and (6)), supporting the proposition that hedge funds prefer firms characterized by overinvestment in R&D. This observation aligns with existing literature suggesting that hedge funds prefer firms that overinvest in R&D due to performance incentives (Fung et al., 2008), high shareholder engagement costs (Gantchev, 2013), and a general preference among institutional investors for firms to focus on their core products and services (Goranova et al., 2007).

Next, we examine non-linearity in the relation between inefficient R&D investment and activists' targeting behaviour. Prior studies document cases of activism that face resistance from both management and blockholders due to managerial incentives and differing beliefs about the value implications of the activist's proposals (Brav et al., 2008; Boyson and Pichler, 2019).¹⁶ We surmise that such resistance is more likely to occur in firms with high levels of inefficient R&D investment since activists seeking to implement their strategy have to make more corrections and thus cause greater “disruption” at these targets, increasing both physical and intangible costs of activism. Results from the quadratic functional form in Panel B show evidence of a curvilinear relation between *R&D Inefficiency* and the likelihood of being targeted, with a turning point for R&D overinvestment at around \$252 million.¹⁷

In summary, our results unveil a strategic pattern in activist targeting, showing a positive and non-linear association between *R&D Inefficiency* and the likelihood of being targeted, particularly for overinvesting firms. Therefore, activists appear to have a strong preference for firms that overinvest in R&D and a strong aversion to firms that underinvest in R&D. Combined, these findings hold significant economic implications, suggesting that activists may be targeting firms with the intention of addressing managerial inefficiencies (Brav et al., 2008; Greenwood and Schor, 2009; Bebchuk et al., 2015). We further investigate this proposition next.

4.2.2. Does the market perceive potential remediation of R&D inefficiency by hedge fund activists in target firms?

Our event study shows the average market-adjusted return around activism announcements is 2.06 percent (t -stat = 6.99) for the 3-day (-1,1) window and 3.73 percent (t -stat = 5.17) for the 21-day (-10,10) window (results not tabulated). These returns are slightly smaller than those documented by prior studies.¹⁸ Table 4 reports the regression analysis of activism announcement returns. Results from the first-stage regressions (column (1)) show the instruments meet both relevance and exogeneity criteria.¹⁹ The baseline results for the full sample in Panel A show no evidence of a significant relation between *R&D Inefficiency*, regardless of how it is measured, and announcement returns.

We further explore whether the relation between activism announcement returns and *R&D Inefficiency* hinges on whether target firms are overinvesting or underinvesting in R&D. Panel B shows a positive association between *R&D Inefficiency* and announcement returns for overinvesting firms. In economic terms, a \$1 million or 1 percent increase in R&D overinvestment is associated with a 0.057 (column (2)) or 0.127 (column (4)) percent increase in announcement returns.²⁰ Results for underinvesting targets are stronger, with Panel C showing a negative relation between *R&D Inefficiency* and announcement returns. Considering that activists typically cut R&D

¹⁶ A case in point is Triun Fund, which targeted Dupont in 2015. The proposal was not supported by Dupont's management and three largest shareholders (BlackRock, State Street, and Vanguard) who collectively owned 16.7 percent of the company's shares. Undeterred, Triun launched a proxy contest to have four of its representatives placed on the board but lost. The company relented and spun off a major division, returning \$9 billion in capital to shareholders.

¹⁷ The turning point is estimated based on column (2) in Panel B of Table 3: $-0.101/(2 \times -0.0002) = \252 million.

¹⁸ For example, the average announcement return is 8.4 percent in Brav et al. (2008) for interventions from 2001 to 2006, and 6 percent in Bebchuk et al. (2015) for interventions from 1994 to 2007.

¹⁹ To save space, only the first-stage regression results for the (-1,1) window for *R&D Inefficiency* (\$) are reported.

²⁰ We also find no evidence of non-linearity in the relation between *R&D Inefficiency* and announcement returns (untabulated).

Table 2
R&D inefficiency at target firms and sample profile.

		Signed R&D Inefficiency (\$)		Signed R&D Inefficiency (%)		Overinvestment (%)
	N	Mean	Median	Mean	Median	
Panel A: Full Sample						
All firms	636	0.123	0.054	0.063	0.046	73.08
Panel B: By industry						
Manufacturing	117	0.122	0.038	0.054	0.024	80.00
Technology	335	−0.005	0.034	0.030	0.038	63.92
Health	184	0.318	0.088	0.133	0.065	85.86
Panel C: By year						
1998	36	0.114	0.047	0.163	0.085	91.11
1999	27	0.162	0.045	0.129	0.032	88.89
2000	25	0.101	0.082	0.139	0.065	87.20
2001	23	0.160	0.044	0.076	0.030	81.74
2002	35	0.064	0.081	0.209	0.101	83.43
2003	35	0.157	0.047	0.063	0.046	83.43
2004	32	0.134	0.050	0.082	0.053	80.63
2005	58	0.153	0.030	0.015	0.015	71.72
2006	72	−0.124	0.059	0.037	0.029	66.94
2007	77	0.002	0.050	0.036	0.037	68.31
2008	79	−0.204	0.037	0.128	0.060	63.54
2009	33	−0.058	−0.001	−0.047	−0.002	49.09
2010	54	0.071	0.047	0.039	0.038	60.00
2011	50	0.585	0.042	0.040	0.029	69.20
Panel D: Descriptive statistics						
	Mean	S.D.		p25	p50	p75
R&D Inefficiency (\$)	0.124	0.953		−0.002	0.054	0.199
R&D Inefficiency (%)	0.063	0.227		−0.002	0.046	0.125
R&D Expenditure (\$mil)	43.986	121.263		4.025	10.591	28.687
R&D Expenditure (%)	0.121	0.165		0.031	0.072	0.138
R&D Productivity (%)	0.084	0.074		0.025	0.053	0.154
Revenue (\$mil)	509.269	1278.636		29.369	100.738	399.197
Market Share (%)	0.312	0.933		0.008	0.032	0.145
Market Cap (\$mil)	732.605	1822.890		53.438	157.483	481.440
PPE (\$mil)	101.919	288.972		3.446	12.390	58.376
CAPX (\$mil)	22.527	63.602		0.903	3.784	13.457
Employees ('000 s)	2.238	5.432		0.159	0.420	1.729
Advertising (\$mil)	7.974	23.550		0.412	1.247	4.745
SGA (\$mil)	142.846	293.448		19.089	45.166	117.503
TFP	2.613	11.675		0.937	1.113	1.390
ROA	−0.028	0.270		−0.075	0.054	0.129
Leverage	0.158	0.258		0.000	0.013	0.242
Payout	0.022	0.082		0.000	0.000	0.000
Cash	0.211	0.193		0.067	0.160	0.290
Return	0.086	0.825		−0.345	−0.089	0.235
B/M	0.627	0.565		0.294	0.511	0.804
Growth	0.212	0.749		−0.054	0.066	0.218
CAPM	0.100	0.055		0.060	0.092	0.132
Tobin's Q	1.960	1.447		1.127	1.555	2.281

This table reports summary statistics on signed R&D inefficiency at R&D firms targeted by hedge fund activists from 1998 to 2011. An R&D firm is one that has reported R&D at least once during the sample period. Panel A reports the statistics of R&D inefficiency for the full sample, and Panels B and C show the partitions by industry and year, respectively. Panel D provides the statistics for test variables. *R&D Inefficiency (\$)* is the absolute dollar difference (\$mil) between actual R&D expenditure and the estimated level of efficient R&D investment; *R&D Inefficiency (%)* is *R&D Inefficiency (\$)* divided by market capitalization; *Overinvestment* is the percentage of firms that are spending more than the (estimated) efficient level; *R&D Productivity* is the output elasticity of R&D; *Revenue* is revenue or sales; *Market Share* is the proportion of sales to total industry sales; *PPE* is net property, plant, and equipment; *CAPX* is capital expenditure; *Employees* is the number of full-time employees; *Advertising* is advertising expenditure; *SGA* is selling, general, and administrative expenditures; *TFP* is total factor productivity, defined as the ratio of sales to the firm's estimated production function; *ROA* is the return on assets, calculated as EBITDA divided by average total assets over the past two years; *Leverage* is long-term debt divided by average book value of assets over the past two years; *Payout* is the sum of dividend and share repurchases divided by EBITDA; *Cash* is total cash divided by average book value of assets over last two years; *Return* is the buy-and-hold return on the firm's stock in the previous year; *B/M* is book value of equity divided by market capitalization; *Growth* is the year-on-year growth in revenue; *CAPM* is the cost of equity, based on beta estimated from the past 252 trading days and a market premium of 8 %; and *Tobin's Q* is market value (market capitalization + total liabilities) divided by replacement value (total assets).

Table 3

2SLS analysis of R&D inefficiency and activists' targeting behaviour.

	R&D Inefficiency (\$)				R&D Inefficiency (%)		
	First-stage	Second-stage			Second-stage		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	All firms	All firms	Over-investing firms	Under-investing firms	All firms	Over-investing firms	Under-investing firms
Panel A: Baseline targeting behaviour							
<i>Assets</i>	38.750*** (13.06)						
<i>Revenue</i>	−0.233 (−0.85)						
<i>R&D Inefficiency</i>		0.024*** (2.86)	0.032*** (2.88)	0.0002 (0.02)	0.076*** (2.91)	0.082*** (3.06)	0.021 (0.24)
<i>R&D Productivity</i>	116.600** (2.45)	−0.076* (−1.86)	−0.058 (−1.21)	−0.05 (−0.34)	−0.077* (−1.97)	−0.062 (−1.34)	−0.069 (−0.46)
<i>Market Share</i>	629.000*** (5.26)	−0.212*** (−2.80)	−0.313*** (−2.63)	−0.009 (−0.09)	−0.065** (−2.54)	−0.078*** (−3.30)	−0.01 (−0.11)
<i>ROA</i>	−82.98*** (−17.31)	0.020** (2.37)	0.027** (2.54)	−0.026 (−1.29)	0.018** (2.31)	0.02** (2.42)	−0.026 (−1.33)
<i>Asset Turnover</i>	29.34*** (11.59)	−0.007** (−2.07)	−0.009** (−2.06)	0.001 (0.15)	−0.004 (−1.44)	−0.004 (−1.33)	−0.0003 (−0.05)
<i>Leverage</i>	−15.93*** (−3.29)	−0.002 (−0.42)	−0.003 (−0.52)	0.009 (0.64)	−0.003 (−0.60)	−0.005 (−0.94)	0.007 (0.46)
<i>Payout</i>	43.57*** (2.83)	−0.026** (−2.47)	−0.051*** (−3.66)	0.003 (0.12)	−0.016* (−1.78)	−0.029*** (−3.14)	0.003 (0.14)
<i>Market Cap</i>	10.72*** (4.28)	−0.012*** (−2.96)	−0.015*** (−2.84)	−0.002 (−0.35)	0.0002 (0.27)	0.001 (1.48)	−0.002 (−1.22)
<i>Cash</i>	−0.197 (−0.05)	0.011* (1.81)	0.011* (1.72)	0.011 (0.70)	0.006 (0.96)	0.006 (0.92)	0.013 (0.80)
		R&D Inefficiency (\$)			R&D Inefficiency (%)		
	First-stage	Second-stage			Second-stage		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	All firms	All firms	Over-investing firms	Under-investing firms	All firms	Over-investing firms	Under-investing firms
<i>Return</i>	−2.191*** (−3.31)	0.003** (2.30)	0.004*** (2.75)	−0.001 (−0.25)	0.002** (2.03)	0.003** (2.26)	−0.001 (−0.22)
<i>B/M</i>	3.619* (1.85)	−0.005 (−1.40)	−0.004 (−1.05)	−0.005 (−0.68)	−0.003 (−1.08)	−0.003 (−0.78)	−0.007 (−0.83)
<i>Tobin's Q</i>	2.007*** (2.79)	−0.001 (−0.71)	0.0003 (0.40)	−0.004** (−2.41)	−0.001** (−1.99)	−0.001 (−1.21)	−0.003*** (−2.66)
<i>Growth</i>	−3.961*** (−6.68)	−0.001 (−1.50)	−0.0001 (−0.12)	−0.008** (−2.17)	−0.002*** (−2.65)	−0.002* (−1.72)	−0.008** (−2.24)
<i>CAPM</i>	−225.0*** (−9.34)	−0.056** (−2.09)	−0.026 (−0.69)	−0.112** (−2.15)	−0.123*** (−6.38)	−0.127*** (−5.92)	−0.114** (−2.33)
<i>Constant</i>	−932.5*** (−25.43)	0.274*** (3.45)	0.342*** (3.21)	0.088 (0.70)	0.043** (2.56)	0.028 (1.35)	0.087** (2.36)
<i>N</i>	27,680	27,680	21,273	6407	27,680	21,273	6407
<i>Adjusted-R²</i>	0.56	0.012	0.013	0.006	0.011	0.013	0.006
<i>F-stat (first stage)</i>	80.24		45.05	56.41		170.18	106.91
<i>Sargan J-statistic</i>		0.15	0.01	1.78	0.36	0.14	2.42

(continued on next page)

Table 3 (continued)

	R&D Inefficiency (\$)				R&D Inefficiency (%)		
	First-stage	Second-stage			Second-stage		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	All firms	All firms	Over-investing firms	Under-investing firms	All firms	Over-investing firms	Under-investing firms
Panel B: Non-linearity in targeting behaviour							
<i>R&D Inefficiency</i>		0.101*** (3.29)	0.149*** (3.20)	0.123** (1.96)	0.764** (2.33)	1.343** (2.61)	1.318 (1.13)
<i>R&D Inefficiency_Squared</i>		−0.0002*** (−3.67)	−0.0002*** (−3.40)	−0.0003** (−2.27)	−0.801** (−2.16)	−1.314** (−2.49)	−3.786 (−1.10)
Controls/Constant		Y	Y	Y	Y	Y	Y
<i>N</i>		27,680	21,273	6407	27,680	21,273	6407
Adjusted- <i>R</i> ²		0.012	0.013	0.006	0.011	0.013	0.006
<i>F</i> -stat (first stage)		68.09	45.75	88.16	156.8	142.91	131
Sargan <i>J</i> -statistic		1.64	1.17	1.55	1.11	0.15	3.06

This table reports the results of IV-2SLS model on activists' targeting behaviour. The dependent variable equals 1 if the firm is targeted by hedge fund activism, and 0 otherwise. *R&D Inefficiency* is instrumented using *Revenue* and *Assets*. Panel A reports the results for tests of a linear association between *R&D Inefficiency* and activists' targeting behaviour, and Panel B for tests of non-linearity in the relation. *R&D Inefficiency (\$)* is the absolute dollar difference between actual R&D expenditure and the estimated level of efficient R&D investment; *R&D Inefficiency (%)* is *R&D Inefficiency (\$)* divided by market capitalization; *R&D Productivity* is the output elasticity of R&D; *Market Share* is the proportion of sales to total industry sales; *ROA* is the return on assets, calculated as EBITDA divided by average total assets over the past two years; *Asset Turnover* is revenue divided by average book value of assets over the past two years; *Leverage* is long-term debt divided by average book value of assets over the past two years; *Payout* is the sum of dividend and share repurchases divided by EBITDA; *Market Cap* is market capitalization; *Cash* is total cash divided by average book value of assets over last two years; *Return* is the buy-and-hold return on the firm's stock in the previous year; *B/M* is book value of equity divided by market capitalization; *Tobin's Q* is market value (market capitalization + total liabilities) divided by replacement value (total assets); *Growth* is the year-on-year growth in revenue; and *CAPM* is the cost of equity, based on beta estimated from the past 252 trading days and a market premium of 8 %. All continuous variables are winsorized at the 1st and 99th percentiles. Industry, year, and industry-year fixed effects are included. Analysis is performed over five imputed datasets and reported coefficients and *t* statistics (in parentheses) are pooled in accordance with Rubin's (1987) rules. Standard errors are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

expenditure in target firms (Brav et al., 2018), our results suggest that activism at underinvesting firms is viewed by the market as value-destroying while activism at overinvesting firms is viewed as value-enhancing. This proposition aligns with the theoretical underpinnings that activists intervene to correct inefficient R&D investment. Our findings are robust to the inclusion of hedge fund fixed effects, which control for the reputation of hedge funds in successfully implementing their strategies, and dummy variables for the activist's stated objectives.

4.2.3. Do hedge funds correct inefficient R&D at target firms?

Table 5 reports results from the DiD analysis of whether hedge fund activists facilitate more efficient R&D practices at target firms. They show reductions in R&D expenditure in target firms following hedge fund activism, consistent with Brav et al. (2018). Panel A indicates that target firms cut R&D expenditure by an average of \$8.43 million (column (1)) or 1.6 percent of market capitalization (column (2)) relative to matched firms in the three years following activism. These figures represent an average reduction in R&D expenditure of \$44 million or 20 percent of market capitalization, respectively. Columns (3) to (5) reveal no evidence of significant changes in *R&D Productivity* and *R&D Inefficiency* following activism. Industry-level analysis in Panels B to D shows that cuts to R&D expenditure are concentrated among manufacturing and technology firms, with no significant changes in inefficient R&D investment across any of the industries. Therefore, despite cuts to R&D expenditure, we find no evidence that activists address inefficient R&D investment at the firms they targeted.

Next, we explore changes in other factors of production, i.e., sales (*Revenue*), capital (*PPE*), labour (number of *Employees*), and *Advertising* expenditure after the intervention. Activists are well known for advocating a wide range of changes at target firms, including cuts to assets (Brav et al., 2008, 2018) and to employees and operating expenditure (DesJardine and Durand, 2020). Brav et al. (2018) note that R&D expenditure is scaled back in proportion to a reduction in assets. Examining other factors of production is thus necessary to determine if the scaling back of R&D expenditure is offset by changes in the scale of the firm and represents a conscious decision to make the firm “leaner” rather than dealing with inefficient investment problems.

Table 6 corroborates existing findings in the literature. Relative to matched firms, target firms experience a decline in revenue (by \$109.8 million in column (1)), *PPE* (by \$29.46 million in column (2)), and the number of employees (by 389 in column (4)) in the three years following activism. Industry-level analysis in Panels B to D show similar results, with changes in factors of production being concentrated among manufacturing and technology firms but not healthcare firms. Relative to matched firms, manufacturing targets experience the greatest impacts from activism, with revenue, *PPE*, and number of employees declining by \$303.7 million, \$89.46 million, and 1225, respectively. In comparison, technology targets experience declines in sales by \$75.51 million and in workforce by 289 employees. Relative to matched firms, there is no evidence of significant changes in advertising expenditure, total factor productivity, or the firm's production inputs in the three years following activism.

To further confirm the above lacklustre response to activism, we test whether changes in innovation and factors of production translate to changes in company fundamental and performance metrics at target firms. We report the results from the DiD analysis in Table 7. Panel A shows that relative to matched firms, target firms experience declines in market share (by 0.074 percent in column (2)), total assets (by \$106.7 million in column (4)), and net profit (by \$20.111 million in column (5)) in the subsequent three years. Hence, it appears that profits are scaled back in proportion to the reduction in total assets, and that activists sell profitable assets. Similarly, columns (6) and (7) show that relative to matched firms, target firms experience a significant cut in capital expenditure (*CAPX*) and selling, general and administrative (*SGA*) expenditure of \$7.95 million and \$25.58 million, respectively, in the three years following activism. Industry-level analysis shows a significant decline in target firms' market share across all industries. While changes in total assets and net profit are concentrated at manufacturing targets (Panel B), changes in capital expenditure are concentrated at technology (Panel C) and healthcare (Panel D) targets.

In sum, our findings indicate that activists prioritize making the firm “leaner” rather than addressing inefficient R&D. Consequently, their investment-limiting interventions neither hinder nor create shareholder value. Our results thus add to the puzzle why activist investors target firms that are not long-term value-enhancing propositions to them (Coffee and Palia, 2016; Cremers et al., 2020; DesJardine and Durand, 2020). Moon (2024) shows that financial conditions shape R&D project choices, with firms under financial strain opting for high-risk, high-reward projects, while those with better conditions choose safer projects with modest returns. This pattern suggests that external pressures, including hedge fund activism, may push firms towards riskier R&D investments for immediate gains, potentially at the expense of sustainable innovation and long-term growth.

A caveat is however in order – our analysis assumes that activist investors have the ability, skill, and expertise to assess the quality of a firm's innovation-related activities. Our results suggest that this might not be the case since activism is costly, implying that the decision to target a firm is likely based on the activist's evaluation of net benefits to mitigating R&D inefficiency. Unlike over-investment in CAPEX and large acquisitions (M&As), R&D expenditures or innovation-related activities are much less visible in real time to outsiders and thus subject to greater information asymmetry. Therefore, activist investors may find it more difficult to assess the quality of a firm's innovation-related activities and to identify R&D investment inefficiencies at target firms.²¹ Additionally, changes in R&D investment may take longer than our three-year observation period to manifest in firm performance metrics. The median holding period of hedge funds for completed deals is one year (Brav et al., 2008), implying that activists are unlikely to “stick around” for innovation efficiency gains.

²¹ The fact that only 23.7 percent activist targets we examine had R&D expenditures before becoming targets suggests that identifying and correcting inefficient R&D may not be a core skill for activist investors.

Table 4
2SLS analysis of the determinants of activism announcement returns.

	R&D Inefficiency (\$)			R&D Inefficiency (%)	
	First-stage	Second-stage		Second-stage	
	(1)	(2)	(3)	(4)	(5)
	CAR(-1,1)	CAR(-1,1)	CAR(-10,10)	CAR(-1,1)	CAR(-10,10)
Panel A: All target firms (N =592)					
Assets	43.11*** (3.74)				
Revenue	0.706 (0.53)				
R&D Inefficiency		-0.0003 (-0.01)	-0.001 (-1.50)	-0.002 (-0.03)	-0.307 (-1.61)
Constant and Controls		Y	Y	Y	Y
Adj. R ²	0.632	0.331	0.389	0.352	0.429
F-stat (first stage)	10.31				
Sargan J-statistic		0.00	0.09	0.01	0.00
Panel B: Overinvesting firms (N = 429)					
R&D Inefficiency		0.057* (1.87)	-0.005 (-0.09)	0.127** (2.48)	-0.048 (-0.44)
Constant and Controls		Y	Y	Y	Y
Adj. R ²		0.39	0.518	0.539	0.524
Panel C: Underinvesting firms (N=163)					
R&D Inefficiency		-0.120*** (-3.66)	-0.135* (-1.84)	-1.258** (-1.96)	-1.402 (-1.14)
Constant and Controls		Y	Y	Y	Y
Adj. R ²		0.62	0.753	0.277	0.64

This table reports IV-2SLS regression results for activism announcement returns. The dependent variable is the 3-day cumulative market-adjusted return (CAR) around the activism announcement in columns (2) and (4), and the 21-day CAR in columns (3) and (5). First-stage results are reported in column (1). *R&D Inefficiency* is instrumented using *Revenue* and *Assets*. *R&D Inefficiency (\$)* is the absolute dollar difference between actual R&D expenditure and the estimated level of efficient R&D investment; and *R&D Inefficiency (%)* is *R&D Inefficiency (\$)* divided by market capitalization. Panels A, B, and C report the results for all target firms, overinvesting targets, and underinvesting targets, respectively. All continuous variables are winsorized at the 1st and 99th percentiles. Industry, year, industry-year, transaction type and hedge fund fixed effects are included. Analysis is performed over five imputed datasets. Reported coefficients and *t* statistics (in parentheses) are pooled in accordance with Rubin's rules. Standard errors are clustered at the hedge fund level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3. Additional analysis

4.3.1. Activists' investment objectives

We test whether activism outcomes differ according to the activist's investment objectives, as stated in Item 4 ("Purpose of Transaction") of the Schedule 13D filing. We divide the sample into two groups: (i) investment targets, where activists acquire a stake for general investment or undervaluation purposes²²; and (ii) agenda targets, where activists state their dissatisfaction with the current management approach and announce a specific agenda to pursue at the target firm, such as making changes to capital structure, payouts, business strategy, governance, and the sale of all or part of the firm. We record 458 investment targets and 174 agenda targets in our sample. Of those with a specific agenda, 46 cite changes to capital structure and payouts, 63 cite changes to business strategy, 72 cite the potential sale or takeover of the target, and 89 cite changes in governance practices.²³

To perform the analysis, we expand Eq. (14) to include two additional interactions with *Investment* and *Agenda*, and estimate the following triple-DiD (DDD) specification:

$$y_{i,t} = \alpha_i + Investment \times [\beta_1 \bullet HFA_i \times Post_{i,t} + \beta_2 Post_{i,t}] + Agenda \times [\beta_3 \bullet HFA_i \times Post_{i,t} + \beta_4 Post_{i,t}] + \gamma_i' x_{i,t} + \varepsilon_{i,t} \quad (15)$$

where *Investment* and *Agenda* are dummies that equal 1 if the stated objective involves investment purposes or a specific agenda, respectively, and 0 otherwise; and *Post* equals 1 if the observation for the target firm (matched firm) is within one to three years after the activism-event (pseudo-event) year, and 0 otherwise. We also code these dummies for the matched firms and report *F*-tests for differences in coefficients within groups ($\beta_1 + \beta_2 = 0$, $\beta_3 + \beta_4 = 0$) and between groups ($\beta_3 - \beta_1 = 0$).

²² This investment objective is more passive, with activists exerting their influence through the threat of activism (Klein and Zur, 2009; Boyson and Pichler, 2019; Gantchev et al., 2019) or a "halo effect" consistent with the monitoring prowess of large shareholders (Cronqvist and Fahlenbrach, 2009). In most cases, the activists explicitly state they are satisfied with management and stand ready to assist or leave open the possibility of becoming active in the future.

²³ The numbers add up to more than 174 since the activist can cite more than one specific agenda.

Table 5
Difference-in-difference (DiD) analysis of post-activism changes in innovation.

	(1) <i>R&D Expenditure</i> (\$)	(2) <i>R&D Expenditure</i> (%)	(3) <i>R&D Productivity</i> (%)	(4) <i>R&D Inefficiency</i> (\$)	(5) <i>R&D Inefficiency</i> (%)
Panel A: Full sample (N = 6687)					
<i>HFA</i> × <i>Post</i>	−8.428*** (−2.68)	−0.016* (−1.81)	0.002 (0.47)	−1.371 (−0.35)	−0.013 (−1.50)
<i>Post</i>	4.846** (2.18)	−0.003 (−0.38)	0.000 (−0.11)	2.143 (0.56)	−0.007 (−0.90)
<i>R</i> ²	0.942	0.606	0.853	0.845	0.591
Panel B: Manufacturing firms (N = 1277)					
<i>HFA</i> × <i>Post</i>	−8.212* (−1.67)	0.009 (0.80)	0.001 (0.92)	−7.771 (−0.92)	0.008 (0.71)
<i>Post</i>	6.606 (1.60)	−0.003 (−0.43)	−0.001 (−1.45)	11.726 (1.07)	−0.003 (−0.33)
<i>R</i> ²	0.962	0.583	0.892	0.825	0.578
Panel C: Technology firms (N = 3477)					
<i>HFA</i> × <i>Post</i>	−7.683* (−1.71)	−0.027** (−2.25)	0.003 (1.20)	3.166 (0.51)	−0.022* (−1.76)
<i>Post</i>	4.540* (1.71)	0.005 (0.44)	−0.001 (−0.44)	−1.166 (−0.21)	−0.003 (−0.33)
<i>R</i> ²	0.947	0.629	0.923	0.838	0.605
Panel D: Healthcare firms (N = 1933)					
<i>HFA</i> × <i>Post</i>	−9.977 (−1.53)	−0.024 (−1.21)	−0.003 (−0.74)	−8.83 (−0.96)	−0.023 (−1.16)
<i>Post</i>	5.014 (0.98)	−0.010 (−0.62)	0 (−0.12)	5.41 (0.53)	−0.01 (−0.59)
<i>R</i> ²	0.931	0.609	0.906	0.885	0.603

This table reports results for DiD analysis of changes in innovation for the following dependent variables: (1) *R&D Expenditure* (\$); (2) *R&D Expenditure* (%), measured as *R&D Expenditure* (\$) divided by market capitalization; (3) *R&D Productivity*, measured as the output elasticity of R&D; (4) *R&D Inefficiency* (\$), measured as the absolute dollar difference between actual R&D expenditure and the estimated level of efficient R&D investment; and (5) *R&D Inefficiency* (%), measured as *R&D Inefficiency* (\$) divided by market capitalization. *HFA* equals 1 if the firm is targeted by an activist, and 0 otherwise. *Post* equals 1 if the observation for the target firm (matched firm) is within one to three years after the activism-event (pseudo-event) year, and 0 otherwise. Results for the full sample are in Panel A, and for manufacturing, technology, and healthcare targets in Panels B, C, and D, respectively. Dependent variables are winsorized at the 1st and 99th percentiles. Firm fixed effects, *Market Cap*, and *Age* are included. Year fixed effects are included in all specifications except for specification (3). Analysis is performed over five imputed datasets. Reported coefficients and *t* statistics (in parentheses) are pooled in accordance with Rubin's rules. Standard errors are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8 shows no evidence of significant changes in the dollar value of *R&D Inefficiency* both between and within the two groups (column (2)), but some evidence of significant changes in the proportion of *R&D Inefficiency* relative to market capitalization (column (3)). However, *F*-tests indicate that these changes may be driven by significant declines in sales revenue, profitability, and market capitalization at agenda targets relative to investment targets ($\beta_3 - \beta_1 = 0$). Relative to matched firms, declines in the various innovation, production, and firm performance measures are significantly greater at agenda targets than at investment targets. For example, we find significant declines in revenue (by \$195.4 million), profitability (by \$57.26 million), and market capitalization (by \$307.40 million) at agenda targets. The decline in profitability at agenda targets occurs despite significant cuts in SGA expenditures (by \$40.32 million in column (8)), employees (by 497 in column (6)), and *PPE* (by \$35.81 million in column (5)). Investment targets experience significant declines in sales revenue (by \$82.25 million), *R&D* expenditure (by \$6.950 million), *PPE* (by \$27.77 million), employees (by 338), and SGA expenditures (by \$19.06 million), but no significant changes in profitability and firm value. Therefore, at investment targets, increases in total factor productivity likely offset the negative impacts of declines in sales and other factors of production on profitability. Our findings are robust to the removal of firms where the activist intends to pursue the sale of all or part of the company (untabulated), highlighting the negative impact of intrusive interventions on firm value and subdued impact of passive monitoring on firm value.

4.3.2. Activists' reputation effects

We further examine the extent to which the reputation and clout of activists may drive activism outcomes. Krishnan et al. (2016) find more positive announcement returns and better long-run outcomes, including *ROA* and *R&D* expenditure, for firms targeted by a top 10 hedge fund activist relative to other activists. Since top 10 hedge funds are larger and older, and have access to more resources to persuade the board to agree to their demands, they may be more effective at creating firm value by reducing inefficient R&D investment. Following Krishnan et al. (2016), we define a top 10 hedge fund as one with a top 10 aggregate dollar investment in R&D firms in the previous three years. Since we require three years of data to compute aggregate investment, our test of the reputational effects of hedge funds is limited to the period 2001–2011. During this period, there were 101 interventions by top 10 hedge fund activists.

We report the results in Panel B of Table 8, where *Top10* takes the value of 1 if the firm has been targeted by a top 10 hedge fund,

Table 6

Difference-in-difference (DiD) analysis of post-activism changes in factors of production.

	(1) <i>Revenue</i> (\$)	(2) <i>PPE</i> (\$)	(3) ϕ (%)	(4) <i>Employees</i>	(5) β (%)	(6) <i>Advertising</i> (\$)	(7) ψ (%)	(8) <i>TPF</i>
Panel A: Full sample (N = 6687)								
<i>HFA</i> × <i>Post</i>	−109.80*** (−3.76)	−29.46*** (−3.19)	0.008 (1.37)	−0.389*** (−3.01)	0.001 (0.18)	−2.11 (−1.31)	0.003 (0.87)	1.445 (1.35)
<i>Post</i>	91.65*** (4.10)	21.55*** (3.34)	−0.017*** (−3.80)	0.329*** (2.83)	0.003 (0.50)	0.902 (0.59)	−0.005** (−2.37)	−1.241 (−1.33)
<i>R</i> ²	0.971	0.967	0.77	0.968	0.85	0.8	0.888	0.546
Panel B: Manufacturing firms (N = 1277)								
<i>HFA</i> × <i>Post</i>	−303.70*** (−2.66)	−89.46*** (−3.08)	−0.004 (−0.68)	−1.225** (−2.52)	0.012 (0.63)	−4.41 (−0.68)	0.005 (0.41)	−0.01 (−0.25)
<i>Post</i>	177.60** (2.16)	67.56*** (2.99)	−0.007 (−0.76)	1.374*** (2.71)	−0.016 (−1.30)	2.791 (0.41)	−0.010*** (2.70)	0.01 (0.28)
<i>R</i> ²	0.976	0.981	0.669	0.98	0.58	0.86	0.571	0.57
Panel C: Technology firms (N = 3477)								
<i>HFA</i> × <i>Post</i>	−75.51** (−2.35)	−12.73 (−1.09)	0.012 (1.37)	−0.289* (−1.75)	−0.01 (−1.62)	−1.573 (−0.71)	0.002 (1.05)	−0.031 (−0.81)
<i>Post</i>	53.39** (2.24)	7.263 (1.00)	−0.017** (−2.43)	0.212 (1.45)	0.012*** (2.65)	0.84 (0.38)	−0.003** (−2.46)	−0.004 (−0.10)
<i>R</i> ²	0.966	0.96	0.753	0.959	0.641	0.741	0.802	0.52
Panel D: Healthcare firms (N = 1933)								
<i>HFA</i> × <i>Post</i>	−56.73 (−1.47)	−14.19 (−1.31)	0.005 (0.59)	−0.064 (−0.78)	0.02 (1.08)	−1.932 (−0.90)	0.004 (0.31)	14.873 (0.29)
<i>Post</i>	49.51 (1.63)	10.35 (1.42)	−0.016** (−2.46)	0.063 (1.05)	−0.006 (−0.40)	0.091 (0.06)	0.001 (0.18)	−9.331 (−0.26)
<i>R</i> ²	0.957	0.942	0.773	0.957	0.714	0.772	0.757	0.415

This table reports the results for DiD analysis of various factors of production in our endogenous growth framework. The dependent variables are (1) *Revenue*; (2) *PPE*; (3) capital productivity (ϕ); (4) *Employees*; (5) labour productivity (β); (6) *Advertising*; (7) advertising productivity (ψ); and (8) *TPF* (total factor productivity). *HFA* equals 1 if the firm is targeted by an activist, and 0 otherwise. *Post* equals 1 if the observation for the target firm (matched firm) is within one to three years after the activism-event (pseudo-event) year, and 0 otherwise. Results for the full sample are in Panel A, and for manufacturing, technology, and healthcare targets in Panels B, C, and D, respectively. Dependent variables are winsorized at the 1st and 99th percentiles. Firm fixed effects, *Market Cap*, and *Age* are included. Year fixed effects are included in all specifications except for specifications (3), (5), and (7). Analysis is performed over five imputed datasets. Reported coefficients and *t* statistics (in parentheses) are pooled in accordance with Rubin's rules. Standard errors are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

and 0 otherwise. Despite reductions in R&D expenditure (column (1)), we find no evidence of significant changes in R&D inefficiency (columns (2) and (3)) within and between the two groups. Only non-top 10 targets experience significant declines in sales revenue, factors of production (as represented by *PPE* and *Employees*), advertising and *SGA* expenditures, profitability, assets, and market capitalization. Restricting the analysis to the top five hedge funds generates similar results (untabulated), highlighting the negative impact of less reputable activists on firm outcomes.

4.3.3. Passive to active stance

As sophisticated investors, activists are likely to target firms whose fundamentals are expected to improve, independent of whether the intervention is due to superior stock-picking abilities or mean reversion. While the propensity-score-matching procedure addresses this selection bias by activists to some extent by including levels and changes in innovation variables, we cannot rule out the possibility that activists' targeting decision is due to unobservable attributes which drive our results.

To disentangle superior stock-picking ability and mean reversion from changes facilitated by activists, we identify cases where the activist switches from a passive to an active stance at the target firm,²⁴ following the method in Brav et al. (2018). The fact that only the activist's stance changes at the switch provides a cleaner identification of the long-term impacts of activism, allowing for stronger causal interpretation. Additionally, using the activist's passive holdings as the control group helps filter out unobserved hedge fund heterogeneity using hedge fund fixed effects.

We collect Schedule 13 G filings lodged with the SEC by hedge fund activists that intervened in our sample of innovative target firms. From these filings, we isolate instances where the activists switch their stance by lodging a Schedule 13D filing, and estimate the following DiD specification:

$$y_{i,t} = \alpha_i + \beta_1 \text{Switch}_{i,t} \times \text{Post}_{i,t} + \beta_2 \text{Post}_{i,t} + \gamma_i' x'_{i,t} + \varepsilon_{i,t}, \quad (16)$$

²⁴ An investor who files a Schedule 13 G form cannot by law take actions that influence firm policies and control, including communicating with management regarding firm strategies. Such an investor who takes a passive stance would also not file a Schedule 13D since it imposes additional legal and disclosure obligations on the investor. There are however some exceptions, such as when the passive investor acquires 20 percent or more of the stock in which the investor loses Schedule 13 G eligibility.

Table 7

Difference-in-difference (DiD) analysis of post-activism changes in company fundamentals and performance metrics.

	(1) <i>Growth</i>	(2) <i>Market Share</i>	(3) <i>ROA</i>	(4) <i>Assets</i>	(5) <i>NPAT</i>	(6) <i>CAPX</i>	(7) <i>SGA</i>	(8) <i>CAPM</i>	(9) <i>Tobin's Q</i>	(10) <i>Market Cap</i>
Panel A: Full sample (N = 6687)										
<i>HFA</i> × <i>Post</i>	−0.034 (−0.87)	−0.074*** (−3.29)	0.01 (0.87)	−106.70** (−2.13)	−20.11*** (−2.99)	−7.950*** (−3.12)	−25.586** (−2.45)	0.24 (0.93)	0.018 (0.22)	−80.91 (−1.25)
<i>Post</i>	0.021 (0.60)	0.024 (1.59)	0.006 (0.53)	105.20*** (2.78)	18.63*** (3.54)	3.617* (1.84)	18.606*** (2.70)	−0.137 (−0.63)	0.117 (1.46)	81.79* (1.65)
<i>R</i> ²	0.269	0.967	0.761	0.961	0.938	0.913	0.964	0.639	0.663	0.925
Panel B: Manufacturing firms (N = 1277)										
<i>HFA</i> × <i>Post</i>	−0.04 (−1.28)	−0.281*** (−2.61)	−0.002 (−0.21)	−347.50** (−2.39)	−40.54* (−1.84)	−13.54 (−1.41)	−35.994** (−2.39)	−0.053 (−0.09)	−0.021 (−0.26)	−7.099 (−0.05)
<i>Post</i>	0.007 (0.26)	0.085 (0.96)	0.006 (0.63)	305.90*** (2.61)	29.14 (1.61)	3.022 (0.28)	31.944*** (2.74)	−0.373 (−0.85)	−0.032 (−0.47)	26.49 (0.24)
<i>R</i> ²	0.335	0.953	0.733	0.976	0.923	0.903	0.98	0.679	0.707	0.899
Panel C: Technology firms (N = 3477)										
<i>HFA</i> × <i>Post</i>	−0.012 (−0.39)	−0.022** (−2.23)	−0.002 (−0.16)	−72.95 (−1.06)	−8.501 (−1.48)	−5.791* (−1.78)	−23.656** (−2.32)	0.041 (0.12)	−0.043 (−0.52)	−57.17 (−0.75)
<i>Post</i>	0 (−0.01)	0.011 (1.44)	0.01 (0.96)	67.42 (1.22)	10.85*** (2.66)	2.295 (1.28)	15.410* (1.94)	−0.178 (−0.62)	0.117 (1.56)	83.36 (1.46)
<i>R</i> ²	0.379	0.963	0.67	0.955	0.953	0.921	0.958	0.678	0.661	0.932
Panel D: Healthcare firms (N = 1933)										
<i>HFA</i> × <i>Post</i>	−0.06 (−0.53)	−0.017** (−2.26)	0.049 (1.51)	−97.17 (−1.33)	−23.81 (−1.54)	−4.549* (−1.71)	−26.68 (−0.47)	0.814* (1.83)	0.155 (0.60)	−206.5 (−1.29)
<i>Post</i>	0.128 (1.15)	0.008 (1.39)	−0.002 (−0.05)	77.07 (1.62)	24.82* (1.95)	1.737 (0.81)	14.174 (0.50)	−0.328 (−0.84)	0.279 (0.92)	122.9 (0.91)
<i>R</i> ²	0.212	0.978	0.766	0.94	0.921	0.893	0.956	0.594	0.614	0.933

This table reports the results for DiD analysis of firm fundamentals and performance measures: (1) *Growth*, the year-on-year growth in revenue; (2) *Market Share*, the proportion of the firm's sales to total industry sales; (3) *ROA*, the return on assets; (4) total *Assets*; (5) *NPAT*, net profit after tax; (6) *CAPX*, capital expenditure; (7) *SGA*, selling, general, and administrative expenditures; (8) *CAPM*, the cost of equity; (9) *Tobin's Q*; and (10) *Market Cap*, market capitalization. *HFA* equals 1 if the firm is targeted by an activist, and 0 otherwise. *Post* equals 1 if the observation for the target firm (matched control firm) is within 1–3 years after the activism year (pseudo-event), and 0 otherwise. Results for the full sample and for subsamples in the manufacturing, technology, and healthcare sectors are in Panels A, B, C, and D, respectively. Dependent variables are winsorized at the 1st and 99th percentiles. Firm and year fixed effects, *Market Cap*, and *Age* are included. Results for Column (2) are reported as percentages. Analysis is performed over five imputed datasets and reported coefficients and *t* statistics (in parentheses) are pooled in accordance with Rubin's rules. Standard errors are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

where $Switch_{it}$ takes the value of 1 if the event is a switch from a passive to active stance, and 0 otherwise; $Post_{it}$ equals 1 if the firm-year observation is within one to three years after the year of the Schedule 13 G filing where there is no subsequent switch (or the year of the Schedule 13D filing if there was a switch), and 0 otherwise; and x'_{it} is the vector of control variables, including market capitalization and firm age. We include year, firm, and hedge fund fixed effects. Overall, there are 109 switching targets.

In Table 9, Panel A shows that despite a cut in R&D expenditure at switching targets of \$7.747 million relative to passive holdings (column (1)), there is no evidence of changes in R&D inefficiency (columns (2) and (3)). Relative to passive holdings, switching firms experience a drop in revenue (by \$71.237 million), number of employees (by 137), SGA expenditures (by \$27.016 million), net profit (by \$11.302 million), total assets (by \$93.597 million), and market capitalization (by \$88.453 million) in the three years following activism. These findings are mostly confirmed in industry-level analysis in Panels B to D, albeit some weak evidence of reduced R&D inefficiency.

4.3.4. Activists' stake in targets

Our final test explores whether activism outcomes are driven by the size of the activist's stake in the target firm. Activists with a larger stake in the target have greater control and voting power, and arguably are more likely to succeed in disciplining firms with high levels of inefficient R&D investment. We divide our sample into four ownership quartiles and adopt a DDD framework similar to Eq. (15), where interaction variables for each of the ownership quartiles are included. We find no evidence of a significant impact of activists' stake on key innovation, operating, and stock performance measures (results not tabulated). The result remains intact when we employ alternative ownership specifications, including coding an interaction for firms in the top percentile of ownership and for firms with above and below median ownership. Thus, our findings support Gantchev's (2013) conjecture that the size of the activist's ownership stake does not influence long-run firm outcomes.

5. Summary and conclusion

We investigate the effectiveness of hedge fund activism in addressing inefficient R&D investment and creating long-term value in target firms. Using an intuitive model of endogenous growth to estimate firm-level R&D efficiency and capture both growth and efficiency channels to profitability and value creation, our results show a positive and non-linear relationship between R&D inefficiency

Table 8

Triple difference-in-difference (DDD) analysis of post-activism changes based on activists' objectives and reputation.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	<i>R&D Expenditure</i> (\$)	<i>R&D Inefficiency</i> (\$)	<i>R&D Inefficiency</i> (%)	<i>Revenue</i>	<i>PPE</i>	<i>Employees</i>	<i>Advertising</i>	<i>SGA</i>	<i>NPAT</i>	<i>Assets</i>	<i>Market Cap</i>
Panel A: Activists' Objectives $y_{i,t} = \alpha_i + \text{Investment} \times [\beta_1 \bullet \text{HFA}_i \times \text{Post}_{i,t} + \beta_2 \text{Post}_{i,t}] + \text{Agenda} \times [\beta_3 \bullet \text{HFA}_i \times \text{Post}_{i,t} + \beta_4 \text{Post}_{i,t}] + \gamma_i \mathbf{X}'_{i,t} + \varepsilon_{i,t}$											
<i>Investment:</i>											
(β_1) <i>HFA</i> × <i>Post</i>	−6.950** (−2.40)	0.348 (0.08)	−0.009 (−0.99)	−82.250*** (−2.61)	−27.77*** (−2.81)	−0.338** (−2.27)	−1.451 (−0.68)	−19.060* (−1.89)	−8.831 (−1.38)	−89.400 (−1.61)	−20.88 (−0.29)
(β_2) <i>Post</i>	4.562** (2.03)	1.333 (0.31)	−0.010 (−1.15)	77.540*** (3.15)	17.88** (2.55)	0.309** (2.36)	0.745 (0.42)	19.218** (2.39)	10.51** (2.05)	73.24 (1.64)	8.093 (0.14)
<i>Agenda:</i>											
(β_3) <i>HFA</i> × <i>Post</i>	−11.502 (−1.26)	−6.348 (−0.74)	−0.026 (−1.39)	−195.400*** (−3.15)	−35.81** (−2.01)	−0.497** (−2.36)	−3.696 (−1.06)	−40.324*** (−2.80)	−57.26*** (−3.33)	−180.300* (−1.83)	−307.400** (−2.33)
(β_4) <i>Post</i>	7.403 (1.13)	5.505 (0.71)	0.003 (0.12)	143.800*** (2.64)	34.90** (2.38)	0.457** (2.32)	1.930 (0.76)	25.258** (2.02)	47.11*** (3.15)	209.300*** (2.67)	325.400*** (2.75)
R ²	0.942	0.846	0.591	0.972	0.967	0.968	0.800	0.967	0.938	0.962	0.925
<i>F-tests:</i>											
$\beta_1 + \beta_2 = 0$	1.59	0.61	6.72**	0.11	3.70*	0.00	2.02	0.36	0.18	0.47	0.13
$\beta_3 + \beta_4 = 0$	0.98	0.00	3.28*	2.95*	0.01	0.09	5.27**	5.76**	1.28	0.25	0.09
$\beta_3 - \beta_1 = 0$	0.65	0.48	0.69	2.77*	0.17	0.49	1.92	2.01	7.16***	0.67	3.69*
Panel B: Activists' Reputation $y_{i,t} = \alpha_i + \text{Top10} \times [\beta_1 \bullet \text{HFA}_i \times \text{Post}_{i,t} + \beta_2 \text{Post}_{i,t}] + \text{Not Top10} \times [\beta_3 \bullet \text{HFA}_i \times \text{Post}_{i,t} + \beta_4 \text{Post}_{i,t}] + \gamma_i \mathbf{X}'_{i,t} + \varepsilon_{i,t}$											
<i>Top 10 Activists:</i>											
(β_1) <i>HFA</i> × <i>Post</i>	−13.202* (−1.81)	3.595 (0.19)	0.017 (0.99)	−28.03 (−0.29)	−29.09 (−0.88)	−0.412 (−1.00)	−1.451 (−0.27)	−2.277 (−0.08)	−8.275 (−0.38)	1.802 (0.01)	134.400 (0.78)
(β_2) <i>Post</i>	12.002** (2.20)	−0.660 (−0.04)	−0.005 (−0.35)	147.400** (2.11)	38.17** (2.10)	0.497** (2.45)	0.790 (0.15)	37.104** (−1.97)	26.78** (2.14)	223.80* (1.90)	83.940 (0.76)
<i>Other Activists:</i>											
(β_3) <i>HFA</i> × <i>Post</i>	−5.738* (−1.81)	−3.148 (−0.61)	−0.021** (−2.17)	−120.400*** (−4.37)	−23.460*** (−3.07)	−0.314** (−2.57)	−1.835* (−1.82)	−28.142*** (−2.84)	−21.10*** (−3.29)	−120.40** (−2.46)	−123.800* (−1.83)
(β_4) <i>Post</i>	2.404 (1.04)	3.432 (0.72)	−0.005 (−0.63)	71.24*** (2.98)	15.61** (2.34)	0.273** (2.21)	0.928 (0.85)	10.549 (1.41)	14.24** (2.38)	69.25* (1.75)	63.670 (1.10)
R ²	0.942	0.845	0.591	0.972	0.967	0.968	0.800	0.964	0.938	0.962	0.925
<i>F-tests:</i>											
$\beta_1 + \beta_2 = 0$	0.19	0.23	1.27	3.83*	0.14	0.06	1.06	5.86**	1.27	5.93**	3.03*
$\beta_3 + \beta_4 = 0$	2.44	0.18	11.54***	8.51***	1.56	0.02	0.94	19.51***	2.57	3.34*	3.12*
$\beta_3 - \beta_1 = 0$	1.35	0.12	3.57*	0.81	0.03	0.08	0.06	1.04	0.30	0.53	1.81

This table reports the results of the triple DiD regression of various innovation, production, and firm performance measures based on the activist objectives (Panel A) and reputation (Panel B). *Investment* equals 1 for targets where activists acquire a stake for general investment or undervaluation purposes, and 0 otherwise; *Agenda* equals 1 for targets where activists announce a specific agenda to be put forward to the target firm's management, and 0 otherwise; *Top 10 (Not Top 10)* equal 1 if the firm has (not) been targeted by a top 10 hedge fund (in terms of aggregate dollar investment in R&D firms in the previous three years), and 0 otherwise; *HFA* equals 1 if the firm is targeted by an activist, and 0 otherwise; *Post* equals 1 if the observation for the target firm (matched control firm) is within one to three years after the activism year (pseudo-event), and 0 otherwise; and $\mathbf{X}'_{i,t}$ represents a vector of control variables (i.e., firm and year fixed effects, *Market Cap*, and *Age*). All dependent variables are winsorized at the 1st and 99th percentiles. Analysis is performed over five imputed datasets and reported coefficients and *t* statistics (in parentheses) are pooled in accordance with Rubin's rules. Standard errors are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 9

Difference-in-difference (DiD) analysis for passive-to-active switching.

	(1) <i>R&D Expenditure</i> (\$)	(2) <i>R&D Inefficiency</i> (\$)	(3) <i>R&D Inefficiency</i> (%)	(4) <i>Revenue</i>	(5) <i>PPE</i>	(6) <i>Employees</i>	(7) <i>Advertising</i>	(8) <i>SGA</i>	(9) <i>NPAT</i>	(10) <i>Assets</i>	(11) <i>Market Cap</i>
Panel A: Full Sample (N = 6735)											
<i>Switch</i> × <i>Post</i>	−7.747*** (−2.86)	−3.969 (−0.61)	−0.022 (−1.20)	−71.237*** (−4.11)	−2.524 (−0.33)	−0.137** (−2.00)	−0.93 (−1.24)	−27.016* (−1.75)	−11.302** (−2.47)	−93.597*** (−3.41)	−88.453** (−2.10)
<i>Post</i>	4.858*** (3.97)	6.092* (1.78)	0.023*** (3.58)	77.813*** (5.77)	9.944*** (3.30)	0.111** (2.05)	0.336 (0.85)	23.481*** (−6.52)	15.719*** (5.31)	106.265*** (4.85)	109.294*** (4.56)
R ²	0.822	0.75	0.407	0.946	0.954	0.951	0.735	0.903	0.869	0.914	0.849
Panel B: Manufacturing firms (N = 825)											
<i>Switch</i> × <i>Post</i>	−2.201 (−0.66)	−7.332 (−0.36)	−0.005 (−0.23)	−164.783** (−2.24)	−16.193 (−0.97)	−0.079 (−0.39)	−1.261 (−0.12)	−20.203*** (−2.63)	−13.979 (0.91)	−113.666 (−1.31)	−52.898 (−0.81)
<i>Post</i>	1.923 (1.14)	5.406 (0.29)	0.012 (1.41)	268.95*** (3.83)	36.133** (2.10)	0.298* (1.68)	0.615 (0.54)	25.516** (2.59)	38.118*** (3.27)	231.522** (2.21)	232.474*** (3.11)
R ²	0.874	0.817	0.492	0.952	0.97	0.987	0.654	0.901	0.857	0.923	0.856
Panel C: Technology firms (N = 2757)											
<i>Switch</i> × <i>Post</i>	−6.235 (−1.24)	6.514 (0.57)	0.002 (0.05)	−34.389 (−1.52)	4.726 (0.51)	−0.133 (−1.57)	−0.202 (−0.11)	−15.276 (−0.94)	−0.061 (−0.01)	−44.591 (−1.41)	−2.55 (−0.04)
<i>Post</i>	1.886 (1.11)	0.401 (0.07)	0.029** (2.01)	41.979** (2.56)	2.825 (1.18)	0.111* (1.80)	−0.625 (−0.68)	11.491*** (2.66)	7.761** (2.54)	47.011** (2.35)	19.753 (0.62)
R ²	0.926	0.761	0.495	0.971	0.957	0.97	0.796	0.956	0.923	0.952	0.868
Panel D: Healthcare firms (N = 3153)											
<i>Switch</i> × <i>Post</i>	−11.013*** (−2.76)	−11.036 (−1.37)	−0.063* (−1.74)	−61.376*** (−3.58)	−5.726 (−0.70)	−0.129 (−1.14)	−1.436** (−2.32)	−35.853 (−0.49)	−9.306 (−1.42)	−79.336** (−2.52)	−105.581 (−1.33)
<i>Post</i>	7.584*** (3.77)	7.960* (1.68)	0.027*** (2.59)	57.339*** (5.13)	6.174 (1.64)	0.076 (0.96)	0.812* (1.67)	29.15*** (4.66)	12.251*** (3.34)	83.174*** (3.99)	119.071*** (3.78)
R ²	0.759	0.728	0.346	0.899	0.857	0.865	0.643	0.857	0.822	0.887	0.864

This table reports the results for DiD analysis of various innovation, production, and firm performance measures, capitalizing on the activist's decision to switch from a passive to an active stance at a target firm (*Switch*). *Post* equals 1 if the observation for the target firm (matched control firm) is within one to three years after the activism year (pseudo-event), and 0 otherwise. Variables are winsorized at the 1st and 99th percentiles. All regression specifications include firm, year, and hedge fund fixed effects, *Market Cap*, and *Age*. Analysis is performed over five imputed datasets and reported coefficients and t statistics (in parentheses) are pooled in accordance with Rubin's rules. Standard errors are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

and the likelihood of a firm being targeted. Activism announcement returns are positively related to R&D inefficiency for firms that overinvest in R&D and negatively related to R&D inefficiency for firms that underinvest in R&D. Considering that activists typically make cuts to R&D expenditures in target firms (Brav et al., 2018), these findings suggest that activism is viewed as value-destroying at underinvesting firms, but value-enhancing at overinvesting firms. Our long-run tests show no significant changes to R&D inefficiency and profitability, with cuts in R&D expenditures being offset by a decline in the scale of the firm. Additional tests show profitability and shareholder value are hindered relative to the activist's passive holdings and for interventions by low reputation activists and by those with a specific corporate agenda.

Overall, our paper raises questions regarding the efficacy of the governance role of activists in dealing with inefficient R&D investment in target firms. It also raises questions as to the motivations for engaging in activism and whether activists derive other pecuniary and non-pecuniary benefits beyond the return on their holdings in the target firm. Such value channels for the activist could include improvements in liquidity, cash flow, and reputation, or supply chain benefits derived from facilitating contractual arrangements with other companies in which the activist owns shares. Alternatively, our finding may well reflect hedge funds' inability to identify R&D investment inefficiencies at target firms since R&D expenditures and other innovation-related activities are subject to a great deal of information asymmetry.

Our research offers valuable insights for a spectrum of stakeholders. For investors, understanding the relationship between activism, R&D efficiency, and firm value becomes paramount, as our results showcase divergent outcomes based on the initial R&D investment status of the targeted firms. Corporations, particularly those susceptible to activist interventions, can glean strategic insights into optimizing their R&D expenditures to navigate these external pressures effectively. Policymakers are prompted to balance shareholder interests with long-term innovation goals, shaping regulatory frameworks to promote economic growth. Our findings also suggest avenues for corporate governance reforms addressing R&D investment inefficiencies, enhancing transparency and accountability within firms. Evidence from Chinese firms shows that over-investment in R&D leads to lower bank loan financing sensitivity and higher trade credit financing sensitivity due to managerial expropriation of resources (Chiu et al., 2024). This phenomenon underscores the broader implications of inefficient R&D investment, emphasizing the need for policies that balance short-term gains with long-term innovation objectives. Understanding the complex relationship between activism and innovation efficiency is crucial for fostering an environment conducive to sustainable economic development and innovation.

Looking forward, the scope for future research extends into refining our understanding of the intricacies surrounding activist interventions in R&D investment by exploring factors such as information asymmetry and activist stock-picking abilities. An avenue for further research involves combining our theoretical framework with Ganchev's (2013) research on the costs of activism to model critical thresholds determining when an activist will target a firm. Such an examination would yield deeper insights into the activist's stock-picking abilities, the costs associated with activism, and potential trading strategies, particularly in light of the positive average abnormal returns observed on the announcement of activism.

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CRedit authorship contribution statement

Justin Watson: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Janice How:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Conceptualization. **Peter Verhoeven:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Conceptualization.

Data availability

The authors do not have permission to share data.

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